

RCF Architectural Amendment: The Three-Layer Regime Protocol

Motivation

The current rewrite evaluates both quantum mechanical structures (superposition, Born rule, particle identity) and general relativistic structures (metric, curvature, Einstein closure) on the same mathematical object — the sector sub-algebra A_k and sub-Hilbert space $H_{\omega,k}$. This forces QM and GR to speak in the same mathematical voice, producing the conflation identified in multiple independent reviews.

This amendment establishes a three-layer architecture that separates these regimes using only objects already present in the framework. No new axioms are required.

The Core Principle

The full Hilbert space carries quantum possibility. Reconciliation sectors carry exact relational structure. Coarse-grained sector metrics carry classical spacetime geometry.

Or equivalently: QM is the algebraic Hilbert-space layer. GR is the coarse-grained metric layer. The reconciliation sector is the bridge between them.

Layer A: Quantum-Universal (QM Domain)

Space: The full kinematic Hilbert space H_{kin} and the full constraint algebra A .

This layer is PERMANENT — never discarded after fracture.

Structures native here: Primitive algebra A and all constraints $\{C_a\}$, Kinematic GNS $(A, \omega_{\text{kin}}) \rightarrow H_{\text{kin}}$, Global Master Constraint M , Obstruction burden $B_{\Delta}[p] = \text{Tr}(pF)$ where $F = \sum \Delta_{ab}^* \Delta_{ab}$, Reconciliation Propagator R_t (gradient descent + dephasing), Full spectral structure of F , Cross-sector coherences $P_i \rho P_j$ ($i \neq j$) — real and dynamically suppressed, not mathematically erased, Sector weights $p_k = \text{Tr}(P_k \rho_{\text{kin}})$, Entanglement and superposition across F -eigenspaces, The exact correlation kernel $K_{\omega}(A,B)$ defined on ALL $A, B \in A$ (not just sector-preserving ones).

Metric: The Exact Metric $d_{\text{ex}}(A,B) = -\ell_0 \log|K_{\omega}(A,B)|$ with full phase information. The triangle inequality may fail for non-commuting observables — the algebraic signature of quantum uncertainty.

Layer B: Sector-Resolved Relational (Exact QM Geometry)

Space: A single reconciliation sector $H_{\omega,k} = P_k H_{\text{kin}}$ with its sub-algebra A_k .

Structures native here: Sector-restricted constraints, Dressed constraint representatives $K_a^{(k)}$, Emergent Dirac bracket $[A,B]_{\text{eff},k}$, Single and Supportive Zero Equivalence, Exact correlation kernel $K_{\omega,k}$ and distance $d_{\omega,k}$, Cubic volume kernel $K_{\omega}(A,B,C)$ and $D=3$ closure, Particles as stable low- B_{Δ} modes, Gauge structure as $\text{Aut}(R_{\text{int}})$, Records: $[R, \rho_k]=0$ observables, Born rule from Z-envariance: $P(\alpha) = \langle \psi | \Pi_{\alpha} | \psi \rangle$ on $H_{\omega,k}$.

Metric: The Exact Sector Metric — preserves algebraic distinguishability. Individual observables are distinguishable. The cubic volume element detects linear dependence. Chirality is meaningful.

Layer C: Coarse-Grained Classical (GR Domain)

Space: The coarse-grained effective manifold M obtained by averaging over F -eigenspaces at scale ϵ .

Structures native here: Smooth 4D Lorentzian manifold $(M, g_{\mu\nu})$, Coarse-Graining Map C_ϵ , Coarse-grained burden tensor $\langle T_{\mu\nu}^A \rangle = \sum_k p_k T_{\mu\nu}^A\{B,(k)\}$ — branch-weighted average, not one sector's private value, Einstein-like closure $G_{\mu\nu} = \kappa_B \langle T_{\mu\nu}^A \rangle$ — valid only at this layer, ADM decomposition, Black hole horizons as coarse-grained causal boundaries, Cosmological expansion, Classical limit $\epsilon \rightarrow 0$.

Metric: The Approximate Metric — smooths exact sector structure. Triangle inequality holds with controlled defect ϵ . Observables effectively commute. Phase information is averaged out.

The Coarse-Graining Bridge: C_ϵ

Definition. Let $F = \sum \Delta_{ab}^* \Delta_{ab}$ have spectral decomposition $F = \sum_k f_k P_k$. The Coarse-Graining Map at resolution scale ϵ projects out all cross-sector coherences that R_t suppresses faster than the observational timescale $1/\epsilon$.

Theorem (Validity). Cross-sector coherences decay as $\exp(-\gamma(f_i - f_j)^2 t)$. The coarse-grained description is exponentially accurate at late times.
 C_ϵ is not an arbitrary smoothing. It is fixed entirely by F and γ — quantities already in RCF. It is the mathematical operation that takes you from the QM domain (Layer B, where sectors are distinguishable) to the GR domain (Layer C, where they blur into a single smooth geometry).

Regime Assignment Table

Structure	Layer	Why
Full A , H_{kin} , R_t , F spectrum	A	QM superposition requires the full space
Cross-sector coherence, entanglement	A	Cannot be defined on a single sector
Sector weights p_k	A	Requires the unfractured ρ_{kin}
Dirac bracket, dressed constraints	B	Defined within a single sector
Exact d_ω , K_ω , $D=3$, chirality	B	Distinguishes algebraic structure

Particles, gauge groups	B	Stable modes within a sector
Born rule, Z-envariance	B	Sectorwise probabilities
Records, measurement	B	Definite values within a sector
Coarse-graining map C_ϵ	$B \rightarrow C$	The QM $\rightarrow$ GR transition
Smooth manifold, $g_{\mu\nu}$	C	Emergent from coarse-graining
Burden tensor $\langle T \rangle = \sum p_k T^\wedge(k)$	C	Branch-weighted average
Einstein closure $G = \kappa \langle T \rangle$	C	Valid only on smooth geometry
ADM lapse/shift	C	Coarse-grained fields
Black hole horizons	C	Coarse-grained causal boundaries
Cosmology	C	Large-scale smoothed structure
Dark matter	C	Cross-sector gravity via p_k

Key Implications

1. Section 2 must be split: exact metric $\rightarrow$ Layer B, approximate metric $\rightarrow$ Layer C.
2. Section 5 must be rewritten: burden tensor is $\langle T \rangle = \sum p_k T^\wedge(k)$, not sector-local.
3. Section 6 must be reframed: horizons are coarse-grained, not exact sector boundaries.
4. Section 8 must be reframed: dark matter is branch-weighted burden, not "other universes."
5. H_{kin} is PERMANENT — never discarded. R_t justifies C_ϵ but does not replace H_{kin} .
6. QM unitarity is restored: exact dynamics on H_{kin} is unitary; R_t is effective open-system dynamics.

The Master Slogan

Exact metric $\rightarrow$ quantum. Approximate metric $\rightarrow$ gravity. Coarse-graining $\rightarrow$ the bridge. The full Hilbert space $\rightarrow$ the permanent foundation of both.

This amendment uses only objects already in the framework ($F, \gamma, K_\omega, d_\omega, C_\epsilon$). It requires zero new axioms.